

Dongho Lee

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Research Interests

Symplectic Geometry, Symplectic Dynamics, Reeb Dynamics, Three-body Problem

Education

Ph.D. in Mathematics

Department of Mathematics, Seoul National University, Seoul, Korea Mar 2018 – Feb 2025

Bachelor of Science in Mathematics

Bachelor of Economics in Economics, Minor in Physics

College of Liberal Studies, SNU Mar 2013 – Feb 2018
 Graduated with Honors

Research Experience

Postdoctoral Researcher

Center for Quantum Structures in Modules and Spaces (QSMS), SNU May 2025 – Present
 – Research on three-body problem and related topics.

Research Institute of Mathematics, SNU Mar 2025 – Apr 2025
 Postdoctoral Supervisor: Prof. Cheol-hyun Cho
 – Research on periodic orbits in the three-body problem via symplectic geometry.

Graduate Student Researcher

Department of Mathematics, SNU Mar 2018 – Feb 2025
 Ph.D Advisor: Prof. Otto van Koert
 – Focus on global hypersurfaces of section and the three-body problem in symplectic and contact geometry.

Publications

- Conley-Zehnder Indices of Spatial Rotating Kepler Problem** arXiv Preprint
 – Description of the moduli space and computing the Conley-Zehnder indices of periodic orbits of spatial rotating Kepler problem.
- Global Hypersurfaces of Section and the Spatial Kepler Problem**
 Advisor: Prof. Otto van Koert Ph.D. Thesis, Dec 2024
- Global Hypersurfaces of Section for Geodesic Flows on Convex Hypersurfaces**
 with Sunghae Cho *Archiv der Mathematik*, 31 Jul 2024
 – Constructed a global hypersurface of section for geodesic flows on convex hypersurfaces with isometric involution, generalizing Birkhoff annuli to higher dimensions.

Talks and Presentations

Sildes and posters are available at my homepage.

Invited Talks

1. **Hamiltonian Dynamics and Three-body Problem**
Rookie's Pitch (at SNU), Seoul, Korea 14 Oct 2025
2. **Three-body Problem and Related Problems**
QSMS 2025 Summer Workshop, Goseong, Korea 20–23 Aug 2025
3. **Periodic Orbits of Rotating Kepler Problem**
Billiards and Quantitative Symplectic Geometry, Helderberg, Germany 14 – 18 Jul 2025
4. **Closed orbits of the spatial rotating Kepler problem**
2025 Korean Mathematical Society Spring Meeting, Daejeon, Korea 24 – 26 Apr 2025

Poster Talks

1. **Periodic Orbits and Symplectic Invariants in the Rotating Kepler Problem**
Celestial Mechanics & N-body Dynamics, Tokyo, Japan 15–16 Nov 2025

Conference Participation

- New Developments in Symplectic Geometry** 25 – 28 Nov 2025
Seoul National University, Seoul, Korea
- East Asian Symplectic Conference 2025** 25 – 29 Sep 2025
Hokkaido University, Sapporo, Japan
- 2025 Symplectic Retreat** 23 – 26 Jan 2025
Stanford Hotel, Tongyeong, Korea
- From Hamiltonian Dynamics to Symplectic Topology and Beyond** 3 – 7 Jun 2024
A 2024 celebration of Claude Viterbo and his mathematics
Institut Henri Poincaré, Paris, France
- Topology of 4-manifolds and Related Topics** 21 – 27 Jan 2024
A conference in honor of Jongil Park's 60th birthday
Ocean Suites Hotel, Jeju, Korea
- East Asian Symplectic Conference 2023** 29 Oct – 4 Nov 2023
Ocean Suites Hotel, Jeju, Korea
- QSMS Workshop on Symplectic Geometry and Related Topics** 5 – 10 Feb 2023
Ocean Suites Hotel, Jeju, Korea
- Workshop on Symplectic Dynamics** 27 Jun – 1 Jul 2022
Instituto Superior Técnico, Lisboa, Portugal
- Symplectic Geometry and Beyond (Part I)** 7 – 10 Feb 2022
Conference on the occasion of Yong-Geun Oh's 61st birthday
Ocean Suite Hotel, Jeju, Korea
- Fukaya 60 Geometry and Everything** 17 – 22 Feb 2019
A conference in honor of Kenji Fukaya's 60th birthday
Kyoto University, Kyoto, Japan

Personal Affiliations

Korean Mathematical Society , Korea	2025 – Present
QSMS-BK21 Symplectic Seminar , Korea	2021 – Present

Teaching Experience

Teaching and Course Assistant , Seoul National University	Mar 2018 – Feb 2025
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Major Courses

Algebraic Topology (Graduate Course)	Fall 2019
Introduction to Topology	Spring 2023, Spring 2020
Introduction to Differential Geometry	Spring, Fall 2021
Selected Topics Seminar (College of Liberal Studies)	Spring 2023
Probability in mathematical and philosophical viewpoints	

General Education Courses

Head Teaching Assistant for Introductory Mathematics Courses	Spring 2020
Assisted in course planning and exam preparation	
Engineering Mathematics	Fall 2024, Fall, Summer, Spring 2023, Fall 2020
Calculus for Life Science	Fall 2022, Spring 2021
Fundamentals and Applications of Mathematics	Spring 2022
Calculus course for liberal arts students	
Calculus Practice	Fall, Spring 2018
Calculus	Fall, Spring 2018

Awards and Honors

Merit-based Scholarship , Department of Mathematics, Seoul National University	2018
National Scholarship , Korean Student Aid Foundation	2015
Merit-based Scholarship , College of Liberal Studies, Seoul National University	2013

References

Prof. Otto van Koert
 Department of Mathematics, Seoul National University
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 Relationship: Ph.D. Advisor

Prof. Cheol-hyun Cho
 Department of Mathematics, POSTECH
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 Relationship: Postdoctoral Supervisor